Assignment 1: Basic SQL

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Div: 1; **Batch**: T1

Statement: Write the simple SQL queries on the given schema.

A. Write the following simple SQL Queries on the University Schema.

- Find the names of all the students whose total credits are greater than 100 .

  Query:

  ```
  SELECT name FROM student WHERE tot_cred > 100;
  ```

  Output:

  ```
  +--------+
  | name   |
  +--------+
  | Zhang  |
  | Chavez |
  | Tanaka |
  +--------+
  ```

- Find the course id and grades of all courses taken by any student named 'Tanaka'.

  Query:

  ```
  SELECT takes.course_id, takes.grade
      -> FROM takes NATURAL JOIN student
      -> WHERE student.name = "Tanaka";
  ```

Output:

```
+-----------+-------+
| course_id | grade |
+-----------+-------+
| BIO-101   | A     |
| BIO-301   | NULL  |
+-----------+-------+
```

- Find the courses which are offered in both 'Fall' and 'Spring' semester (not necessarily in the same year).

  Query:

```
SELECT course.course_id, course.title
    -> FROM course INNER JOIN section
    -> ON course.course_id = section.course_id
    -> WHERE semester = "Fall" OR semester = "Spring"
    -> GROUP BY course_id
    -> HAVING COUNT(DISTINCT semester) = 2;
```

  Output:

```
+-----------+---------------------------+
| course_id | title                     |
+-----------+---------------------------+
| CS-101    | Intro. to Computer Science |
+-----------+---------------------------+
```

- Find the names of all the instructors from Comp. Sci. department.

  Query:

```
SELECT name
    -> FROM instructor
    -> WHERE dept_name = "Comp. Sci.";
```

Output:

```
+------------+
| name       |
+------------+
| Srinivasan |
| Katz       |
| Brandt     |
+------------+
```

- Find the course id and titles of all courses taught by an instructor named 'Srinivasan'

  Query:

```
SELECT teaches.course_id, course.title
    -> FROM teaches NATURAL JOIN course NATURAL JOIN instructor
    -> WHERE instructor.name = "Srinivasan";
```

  Output:

```
+-----------+----------------------------+
| course_id | title                      |
+-----------+----------------------------+
| CS-101    | Intro. to Computer Science |
| CS-315    | Robotics                   |
| CS-347    | Database System Concepts   |
+-----------+----------------------------+
```

- Find the names of instructors who have taught at least one course in Spring 2009.

  Query:

```
SELECT DISTINCT(instructor.name)
    -> FROM instructor NATURAL JOIN teaches
    -> WHERE semester = "Spring" AND year = 2009;
```

Output:

```
+--------+
| name   |
+--------+
| Brandt |
| Kim    |
+--------+
```

B. Create the Railway schema using the commands in the Railway DDL script; The DDL script also contains a description of the railway Schema. Insert sample data using the command in the file SampleRailwayData.

Write the following Queries for Railway Schema.

- Find pairs of stations (station codes) that have a track (direct connection) with distance less than 20Kms between them.

  Query:

  ```
  SELECT stcode1, stcode2
      -> FROM track
      -> WHERE distance < 20;
  ```

  Output:

  ```
  +---------+---------+
  | stcode1 | stcode2 |
  +---------+---------+
  | BYC     | DR      |
  | BYC     | KRL     |
  | CST     | BYC     |
  | CST     | DR      |
  | CST     | KRL     |
  | GRP     | TNA     |
  +---------+---------+
  ```

- Find the IDs of all the trains which have a stop at THANE

  Query:

  ```
  SELECT trainhalts.id
      -> FROM trainhalts NATURAL JOIN station
      -> WHERE station.name = "THANE";
  ```

Output:

```
+------+
| id   |
+------+
| A65  |
| KP11 |
+------+
```

- Find the names of all trains that start at MUMBAI.

  Query:

```
SELECT train.name
    -> FROM train NATURAL JOIN trainhalts
    -> INNER JOIN station ON trainhalts.stcode = station.stcode
    -> WHERE trainhalts.seqno = 0 AND station.name = "MUMBAI";
```

  Output:

```
+---------------+
| name          |
+---------------+
| CST-AMR_LOCAL |
| CST-KYN       |
+---------------+
```

- List all the stations in order of visit by the train 'CST-AMR_LOCAL'.

  Query:

```
SELECT station.name
    -> FROM train NATURAL JOIN trainhalts
    -> INNER JOIN station ON trainhalts.stcode = station.stcode
    -> WHERE train.name = "CST-AMR_LOCAL"
    -> ORDER BY trainhalts.seqno;
```

Output:

```
+-----------+
| name      |
+-----------+
| MUMBAI    |
| BYCULLA   |
| DADAR     |
| KURLA     |
| GHATKOPAR |
| THANE     |
| DOMBIVALI |
| KALYAN    |
| AMBARNATH |
+-----------+
```